

Model Trust Card

ML & AI Nexus 2026, Round 1 Kaggle Challenge. 30 day unplanned hospital readmission.

Field	Value
Team name	Data Lab
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Final Kaggle submission	submission_13_core3_honest.csv

1. Model summary

Final model is a convex blend of three families, which we call Core 3: a natural B spline GAM (SplineTransformer, 5 knots, cubic, feeding an L2 logistic regression at C=0.1), an Explainable Boosting Machine, and CatBoost. Each family is trained under 5 fold stratified CV and seed bagged across seeds 42, 7 and 123, so every row prediction is the mean of three out-of-fold estimates. Blend weights are non negative and sum to one, fitted by SLSQP.

Preprocessing: missingness indicator flags for the six labs and vitals that can be absent, an n_missing count, median imputation using train medians only, and a small set of engineered clinical features (high_utilizer for prior_admissions_12m >= 3, risk_load, age_x_comorbidity, anaemia and elevated creatinine flags). Categoricals go to CatBoost as native categories and are one hot encoded for the spline GAM.

LightGBM was trained as a fourth family and then deliberately removed. Under nested CV it was the weakest solo model, and adding it to the blend made the honest score worse, so it is documented here as a rejected candidate rather than quietly dropped.

Why this model was selected: it had the best honest CV score. We want to be straight about the size of that win, and section 3 covers it.

2. Validation design

Everything is scored out-of-fold. Level 1 models use 5 fold StratifiedKFold repeated over three seeds. The blend weights are the part that is easy to get wrong, because weights fitted on the same out-of-fold matrix they are scored on will look better than they are. So the weights are refitted inside a separate 5-fold meta split (seed 202) and each row is scored only by weights that never saw it.

With 7000 rows and 881 positives, a single split moves around too much to settle anything. Where we claim one model beats another we use a paired bootstrap over 10000 resamples rather than comparing two point estimates.

We did not tune against the public leaderboard. It is roughly 30 percent of the test set, about 900 rows, and we simulated that the paired difference between our top two submissions has a standard deviation of about 0.00025 on a slice that size. The gap we actually observed between them was 0.00009, so the leaderboard cannot separate them, and we didnt let it try.

3. Predictive performance

All figures below are honest out-of-fold on the 7000 training rows, not a single lucky split.

Model	LogLoss	ROC AUC	Role
Constant 0.125857	0.37844	0.500	Prevalence floor
Logistic regression, C=0.1	0.35261	0.68529	Interpretable reference
LightGBM (seed bagged)	0.35124	0.68851	Rejected, hurts the blend
Spline GAM (seed bagged)	0.34991	0.69023	Core 3 member
CatBoost (seed bagged)	0.34965	0.69226	Core 3 member
EBM (seed bagged)	0.34928	0.69377	Best single model
Four model stack	0.34912		Worse than Core 3
Core 3 (final submission)	0.34855	0.69458	Submitted

Core 3 Brier score is 0.10207, against 0.11002 for the constant and 0.10318 for the logistic regression.

At a threshold equal to the base rate (0.1259) sensitivity is 0.594 and specificity is 0.693, flagging 34.3 percent of patients at a precision of 0.218. Moving the threshold to 0.20 flags 14.0 percent, with sensitivity 0.347, specificity 0.889 and precision 0.311.

The result we think matters most. Core 3 beats a single EBM by 0.00074, and the paired bootstrap puts the 95 percent interval at [-0.00178, +0.00028]. That interval contains zero. We cannot show the three-model ensemble is better than a single glass-box model. The SLSQP optimiser also converges after a single iteration to weights of exactly 0.333, 0.333 and 0.333, and a grid search finds the best possible weights improve on a plain average by 0.000011. The super learner is an average, and the loss surface over the weight simplex is flat.

4. Calibration

No post hoc calibration is applied to the final model, because it did not need any. The calibration slope is 1.0416 and the intercept is 0.0764, both close to the ideal 1 and 0. Expected calibration error over five quantile bins is 0.00677.

Bin	n	Mean predicted	Observed rate
1 (lowest)	1400	0.0517	0.0479
2	1400	0.0761	0.0814
3	1400	0.1009	0.0957
4	1400	0.1380	0.1307
5 (highest)	1400	0.2614	0.2736

We did test recalibration earlier on the LightGBM challenger. Platt scaling bought 0.00007, which is nothing, and isotonic regression actively hurt (0.35590 against 0.35348 uncalibrated). Isotonic wants far more than 881 events. That evidence is why the final model ships uncalibrated.

5. Robustness and stability

What might change between development and deployment: the case mix. Our EDA found the test set carries more Rural (17.5 to 22.5 percent), more District hospital (18 to 23 percent), fewer Teaching hospital (44 to 38.5 percent) and more care_pathway P4 (13.3 to 17.8 percent) patients than training. Missingness is also higher in test for all six affected columns, roughly 1.4 times the training rate.

To find out whether that adds up to a real shift we ran adversarial validation: train a LightGBM to tell a train row from a test row. Out of fold AUC came out at 0.5550. Mild, but above chance, so the shift is real and not just column-by-column noise. The features driving separability were age, length_of_stay_days, n_missing, care_pathway and hospital_type, which lines up with what the EDA had already flagged.

Sensitivity tests run: importance sampling reweighting by the adversarial odds, log transforms of creatinine and length of stay, an extra missingness stress test, and an ablation of care_pathway. care_pathway is the one variable the data dictionary warns is unstable, and its ablation delta is 0.00065 with a standard deviation of 0.00077. That difference is smaller than its own noise, so we keep the variable but do not rely on it, and a deployment that saw pathway codes change could drop it at almost no cost.

6. Subgroup reliability

Skill is the log loss saved against a constant set to that group own observed rate, so a positive number means the model adds something inside the group rather than just tracking the group base rate. Intervals are Wilson 95 percent on the observed rate.

Group	n	Events	Observed (95% CI)	Mean pred.	Skill
Female	3543	439	0.1239 (0.1135 to 0.1352)	0.1239	+0.0308
Male	3457	442	0.1279 (0.1171 to 0.1394)	0.1273	+0.0289
Rural	1243	194	0.1561 (0.1370 to 0.1773)	0.1515	+0.0395

Group	n	Events	Observed (95% CI)	Mean pred.	Skill
Semi urban	1870	204	0.1091 (0.0958 to 0.1240)	0.1145	+0.0290
Urban	3887	483	0.1243 (0.1143 to 0.1350)	0.1227	+0.0254
Age under 30	343	13	0.0379 (0.0223 to 0.0638)	0.0542	-0.0066
Age 30 to 50	2030	159	0.0783 (0.0674 to 0.0908)	0.0758	+0.0069
Age 50 to 70	3150	380	0.1206 (0.1097 to 0.1325)	0.1234	+0.0161
Age 70 plus	1477	329	0.2227 (0.2023 to 0.2447)	0.2153	+0.0363
District	1285	159	0.1237 (0.1068 to 0.1429)	0.1272	+0.0357
General	2623	320	0.1220 (0.1100 to 0.1351)	0.1228	+0.0309
Teaching	3092	402	0.1300 (0.1186 to 0.1423)	0.1274	+0.0265

Sex shows no gap worth acting on. The two Wilson intervals overlap almost completely, and predicted means sit on the observed rates. Rural patients do carry a higher observed rate, and the model tracks it (0.1515 predicted against 0.1561 observed), so the gap is in the population, not introduced by the model.

The under-30 group is where this breaks. Skill is negative, meaning we do worse than simply predicting that group own base rate, and the model overpredicts (0.0542 against 0.0379 observed). With 343 patients and 13 events the interval is wide, so we are not claiming a precise effect size, only that we cannot show any benefit there. These are associations measured on synthetic data. None of it should be read as a causal statement about why a group readmits more.

7. Uncertainty and human referral

We rank patients by the entropy of the predicted probability, which is highest near 0.5, and refer the most uncertain 10 percent to a human. Log loss on the retained 90 percent falls to 0.31756 from 0.34855 overall.

That number on its own would be misleading, because log loss is largest near 0.5 and removing any 10 percent of rows helps a bit mechanically. So we ran a control: referring a random 10 percent, 200 times, gives 0.34857 with a standard deviation of 0.00241, essentially no change. The entropy rule is doing real work.

The clinically useful part: the referred 10 percent has an event rate of 0.3414 and contains 239 of the 881 readmissions, so 27.1 percent of all true readmissions land in the queue a clinician actually looks at.

8. Explainability

The strongest single model in the blend is an EBM, which is a glass box by construction, so every term can be plotted and read directly instead of being approximated after the fact. Across the spline GAM, the EBM and the logistic reference the same predictors dominate: age, prior_admissions_12m and comorbidity_count. On the logistic reference the odds ratios per one standard deviation are 1.58 for age, 1.39 for prior admissions and 1.24 for comorbidity count, and discharge to home carries 0.685.

High risk case (training row 1820, predicted 0.778, was readmitted): age 78, four prior admissions in 12 months, five comorbidities, seven discharge medications, follow up booked at 18.6 days. Cohort medians are 57 years, one prior admission, one comorbidity and 9.9 days to follow up, so the two variables carrying most of this prediction are prior admissions and age, with the late follow up adding to it.

Low risk case (training row 6827, predicted 0.025, not readmitted): age 20.3, no prior admissions, one comorbidity, follow up at 2.9 days. Every driver points the same way.

These are associations. We cannot say that moving a patient follow up from 18.6 days to 2.9 days would lower their risk. Follow up timing on this data is as likely to be a marker of how unwell someone was and how easily they can reach a clinic as it is a lever anybody can pull.

9. Failure modes and limitations

1. Patients under 30. Negative skill, over prediction, 13 events. The model should not be used unsupervised on this group and we would route them to a local base rate or to a human.
2. Transportability. Adversarial AUC of 0.555 says train and test genuinely differ, and the private leaderboard is the other 70 percent of test. Our CV says 0.349 while the public leaderboard says 0.334, so the public slice is easier than our folds. We expect the private ranking to move and would not be surprised by it.
3. The ensemble is not demonstrably better than one EBM. Anyone who deploys three model families believing they bought accuracy takes on three times the maintenance, three times the retraining surface and a much harder explanation, for a difference whose confidence interval contains zero.
4. Missingness is load bearing. We feed missingness flags as features because missingness itself predicts the outcome here, and test missingness already runs about 1.4 times training. A site that imputes labs upstream before the data reaches us would silently delete a signal the the model is using, and nothing in the pipeline would raise an error.

10. Deployment recommendation

Requires additional model development.

An AUC near 0.695 supports triage, not decisions, and calibration alone is not enough to deploy on. Two things block us. The model is worse than a local constant for under 30s, and we cannot show the submitted ensemble beats a single interpretable model, which means the architecture has not earned its complexity. We would go back, fix the under 30 behaviour, and put the EBM forward on its own. If a limited prospective pilot were required sooner, it should exclude under 30s and keep the 10 percent referral queue.

11. Reproducibility

Software: Python 3.14, pandas 3.0.5, numpy 2.5.3, scikit learn 1.9.1, scipy 1.18.1, lightgbm 4.7.0, catboost 1.2.10, interpret 0.7.8 for the EBM. Environment pinned with a pixi lockfile on osx arm64. Parts of the audit were rerun on Fedora under Anaconda Python 3.13 with numpy 2.3.5 and scikit learn 1.9.0, and reproduced the same figures.

Random seeds: model and CV seeds 42, 7 and 123 for seed bagging, meta CV seed 202 for the blend weights, seed 42 for adversarial validation.

Approximate training time: [team to confirm] for the full seed bagged pipeline end to end.

AI assistant use, as the rules require us to declare it. Qwen Coder was used by one of us to draft Python scripts and to talk through which direction to take next when a result was ambiguous. Claude was used by another member for some of the pipeline scripts. Every number in this card was produced by running our own code, and we checked the validation design and the statistical claims ourselves rather than taking them on trust.

12. Conclusion

We trust this model only when it is used to rank patients aged 30 and over for follow up rather than to decide anything on its own, when the most uncertain 10 percent goes to a clinician, and when somebody is watching the case mix for the drift our adversarial check already found at AUC 0.555.